

Prince Zyrus Natividad

Builder & Opinion Editor — Embedded systems · Local AI · Algorithmic trading · Journalism

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[01] EDUCATION

Dominican College of Tarlac
Grade 11, Accountancy, Business & Management (ABM)
2026-2027 · ENROLLED

Capas National High School
Grade 10, Science, Technology & Engineering (STE)
COMPLETED 2026

Self-study: Philippine law, finance, electronics & embedded systems

[02] RECOGNITION

DepEd EDUKAMPYON: Gawad Parangal NATIONAL

1st Runner-Up, National Robotics Competition NATIONAL

MSSPC · 1st Runner-Up, Collaborative Writing
MSSPC · 3rd Place, Editorial Writing
MSSPC · 8th Place, Column Writing
DSPC Qualifier, Editorial Writing
Rotary Regional Training
PARTICIPANT

[03] SKILLS

HARDWARE & EMBEDDED
ESP32, ESP32-CAM, Arduino Uno/Nano, ESP8266, Raspberry Pi, LoRa, sensors, relays, motors, OLED, circuit design, hardware repair

SOFTWARE & AI
Python, Arduino C++, shell scripting, Ollama, LM Studio, LiteLLM, SiliconFlow; Qwen, DeepSeek, Gemma

SYSTEMS
Linux (Arch, Debian, CachyOS), Hyprrland, WSL2, Docker, QEMU, Git/GitHub, Vercel, Cloudflare Tunnels

CREATIVE & COMMS
Editorial and IMRAD writing, DaVinci Resolve, OBS Studio, Figma, Canva, Photoshop, Lightroom, BlenderKit

Languages: English, Filipino

Beyond work: rhythm games, J-pop and Vocaloid, hardware tinkering

[04] PROFILE

Independent developer and Grade 11 ABM student who builds resilient hardware and AI systems, then writes about the world they operate in. Completed a working computer-vision flood-monitoring prototype and a wearable assistive device. Opinion Editor of a campus publication and a national DepEd awardee. Seeking funding or scholarship support for the flood-monitoring prototype.

[05] SELECTED PROJECTS

Non-Contact Flood Monitoring System JAN-MAR 2026 · REF FLDMN-01
Capstone research prototype · computer vision + decentralized alerts

- Measures water level with computer vision, without touching the water.
- Multi-path alerts over BitChat/Bluetooth mesh, SMS and Telegram, so warnings can still reach people when standard networks fail.
- Tested accuracy and alert delivery; documented as an IMRAD research paper.

Hat-a-See · Wearable Assistive Hat REF HATSEE-02
For visually impaired users · Arduino UNO, HC-SR04 ultrasonic sensors, vibration motors, servo

- Engineered distance-based vibration feedback, with pulse rate driven by obstacle distance.
- Fixed ultrasonic crosstalk by firing sensors in sequence; tuned a step-scan servo with hardware-tested pulse widths.
- Delivered firmware, wiring plans, a user guide and a 9-slide presentation.

Quantitative Trading Pipeline REF TRADE-03
Nine-notebook pipeline (NBO-NB8) following López de Prado methodology · in final testing

- Implemented fractional differentiation, triple-barrier labeling, meta-labeling, purged cross-validation, CPCV and the Deflated Sharpe Ratio.
- Designed a five-layer multi-agent system for live micro-capital execution on Alpaca.

[06] MORE BUILDS

LAUNDRY-SAVER FEB-MAR 2026
Rain-adaptive laundry protection on Arduino Nano and ESP8266 with a rain sensor, relay and gear motor; open-source schematics.

Payroll System JUL 2026
Philippine payroll for a 15-employee business: six position levels, TRAIN-law withholding, SSS, PhilHealth and Pag-IBIG.

ESP32 OLED Display AUG 2026
Real-time time-and-icon display programmed and debugged on ESP32 with Arduino IDE.

fzz2md.py T00L
Command-line tool that converts Fritzing .fzz files into Markdown documentation.

Mung Bean Growth Study AUG 2025
STE experiment on the effect of coconut husk ash on plant height growth.

[07] LEADERSHIP & WRITING

Opinion Editor (current) · Managing Editor (previous) CAMPUS PUBLICATION

- Main opinion writer; covers environmental, economic and geopolitical issues for the campus paper.
- Managing Editor of the collaborative-writing team that placed 1st Runner-Up at MSSPC; individually placed 3rd in Editorial and 8th in Column Writing, and qualified for DSPC in Editorial.
- Featured column: *Questions Behind the Chips: The Pax Silica Microchip Initiative in Capas*.
- Designed and ran a structured 44-day journalism and English writing curriculum for his younger sister.



SCAN FOR
THE LIVE PORTFOLIO
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